Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 25)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 The kinetic energy available to a wind turbine is proportional to the cube of the wind velocity. Table 1 shows mean wind velocities at four possible wind farm sites.

Table 1

Site	Mean wind velocity / m s^{-1}	Distance to grid / km	Other features
A: upland ridge	9	12	Migrating bird route; peat bog
B: coastal farmland	8	3	Near village
C: offshore shallow sea	10	15	Sandbanks
D: inland plain	6	2	Arable farmland

0 1 . 1 Calculate how many times more kinetic energy is available at site C than at site D.

[2 marks]

Answer ____ times

0 1 . 2 Use Table 1 to evaluate the four sites for a large wind farm.

[5 marks]

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0 1 . 3 Give **two** advantages of vertical axis wind turbines (VAWTs) over horizontal axis turbines (HAWTs).

[2 marks]

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0 1 . 4 Explain how direct-drive turbines improve on turbines with a gearbox.

[1 mark]

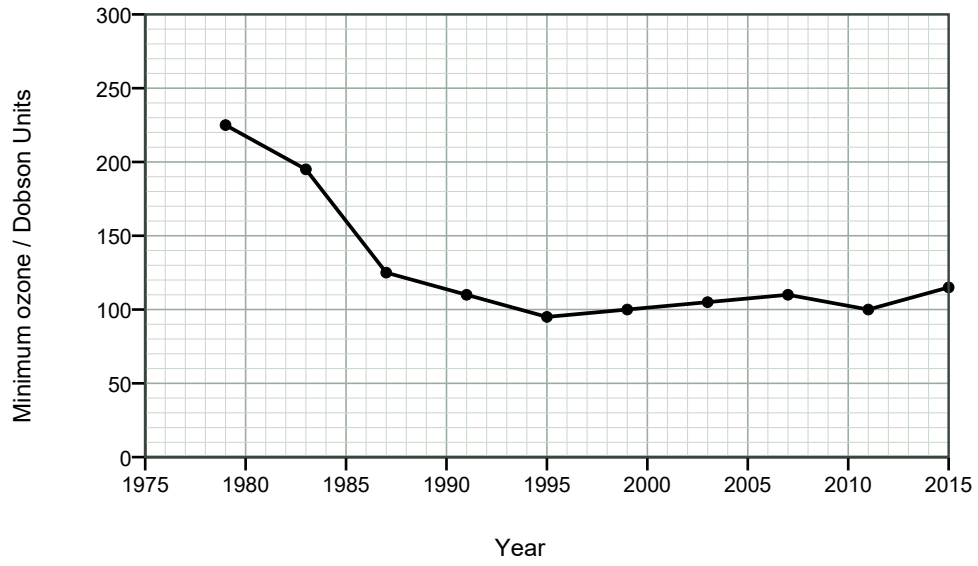
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0 2 Figure 1 shows the minimum ozone concentration recorded over Antarctica each spring between 1979 and 2015.

Figure 1



0 2 . 1 Describe the trend shown in Figure 1.

[2 marks]

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0 2 . 2 Calculate the percentage decrease in minimum ozone between 1979 and 1995.

[2 marks]

Answer ☐ %

0 2 . 3 Explain why ozone depletion is most severe over Antarctica in spring.

[4 marks]

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0 2 . 4 Explain why the data in Figure 1 alone could not confirm that CFCs were the cause of the depletion, and describe how the cause was confirmed.

[2 marks]

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0 3 . 1 Explain why phosphorus is more likely than nitrogen to be a limiting factor on biological productivity.

[3 marks]

0 3 . 2 Explain how mycorrhizal fungi increase phosphate uptake by plants.

[2 marks]

0 3 . 3 Describe how human activities have altered the phosphorus cycle.

[3 marks]

0 3 . 4 Explain **two** strategies for the sustainable management of phosphorus supplies.

[2 marks]

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0 4 . 1 Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

[3 marks]

0 4 . 2 Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

0 4 . 3 Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

0 4 . 4 Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

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0 5 Figure 1 shows the mean concentration of lead in the blood of children in a large city between 1976 and 2000, and the year in which lead additives in petrol began to be phased out.

Figure 1



0 5 . 1 Calculate the percentage decrease in mean blood lead between 1976 and 2000.

[2 marks]

Answer ..... %

0 5 . 2 Explain why tetraethyl lead was added to petrol and why this caused a public health threat.

[3 marks]

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0 5 . 3 Suggest **two** reasons, other than petrol additives, why blood lead concentrations continued to fall after 1990.

[2 marks]

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0 5 . 4 Explain how adding small amounts of phosphoric acid to a water supply reduces lead in drinking water.

[2 marks]

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0 5 . 5 Explain why discarded lead fishing weights and shotgun pellets can poison waterfowl.

[1 mark]

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0 6 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☐

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☐

D To measure the concentration of a pollutant at the point of release ☐

0 6 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 6 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

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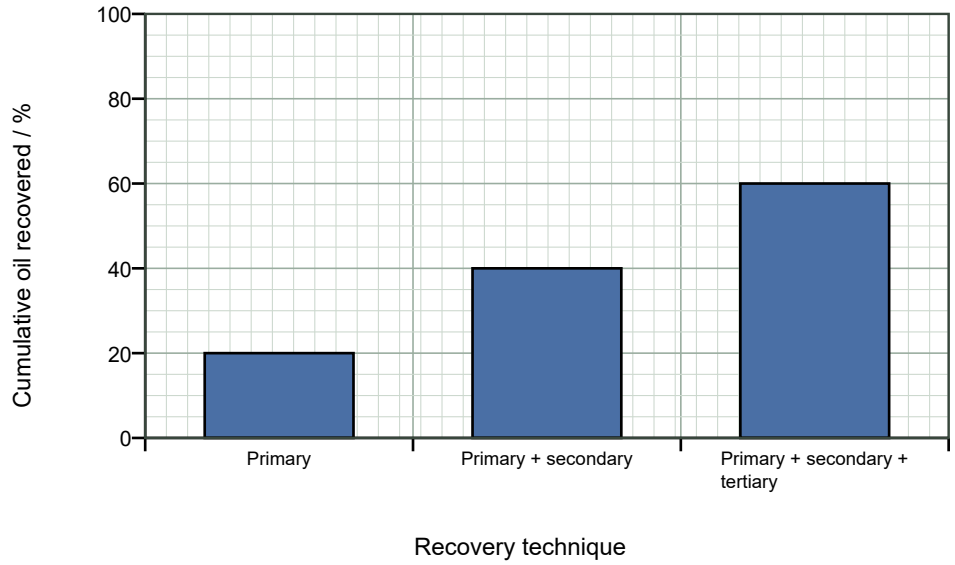
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0 7 Figure 1 shows the percentage of the oil in a reservoir that can be recovered using different recovery techniques.

Figure 1



0 7 . 1 Describe how primary oil recovery works and explain why it recovers only about 20% of the oil.

[2 marks]

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0 7 . 2 Describe secondary recovery and explain how it can be linked to carbon capture and storage.

[2 marks]

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0 7 . 3 Describe **three** tertiary (enhanced) recovery techniques.

[3 marks]

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0 7 . 4 Explain **three** advantages of directional drilling.

[3 marks]

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0 8 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 8 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 8 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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0 9 . 1 Explain how hydrogen can be used to store energy from intermittent renewable resources, and describe **two** ways the stored energy can be released.

[4 marks]

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0 9 . 2 Compare **three** methods of storing hydrogen, giving a disadvantage of each.

[3 marks]

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0 9 . 3 Explain how a fuel cell differs from a rechargeable battery.

[2 marks]

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0 9 . 4 Explain what is meant by the 'hydrogen economy'.

[1 mark]

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1 0 Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O_2)	2
Triatomic oxygen / ozone (O_3)	3

1 0 . 1 Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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1 0 . 2 Give **two** other ways in which the atmosphere supports life.

[2 marks]

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1 0 . 3 Explain why the atmosphere is thermally stratified into layers.

[1 mark]

1 0 . 4 The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

OR

1 1 . 2 Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS